Learning the Meanings of Justice

A proposed collaboration in language representation learning and transitional justice

An ongoing Columbia research program offers an opportunity to develop language models that capture how people express justice, harm and reparation. The team includes **Peter Dixon**, Associate Professor of Practice and Director of Columbia's M.S. in Negotiation and Conflict Resolution, and Senior Fellow at the Center for Justice.[1] Its Spanish-language and transitional-justice expertise could support a collaboration grounded in Colombia's Special Jurisdiction for Peace (JEP).

Exploratory analysis. I built and ran a reproducible pilot testing whether language representations distinguish a harm's initiating event from its consequences. Across 24 constructed comparisons, generating an aspect-specific answer before embedding increased Qwen's accuracy on both requested distinctions from 42% to 75%. AI judges consistently endorsed both intended answers on seven comparisons, motivating expert annotation to clarify the distinctions and establish a reliable evaluation set.

The technical question. Can domain-informed human supervision improve how encoders represent specific aspects of meaning across new accounts? An encoder maps a passage to a vector used for comparison, retrieval and clustering. Conditional semantic similarity asks how that comparison should change with the question being asked. Two narratives might share an initiating event while describing different consequences. C-STS formalizes conditional similarity [1]; Tu et al. generate aspect-specific answers before comparison [2]; CASE learns a compact transformation of conditional embeddings [3]. These provide concrete baselines for testing the value of expert supervision.

A shared research opportunity. Columbia's School of Professional Studies (SPS) team reports a corpus of **243 hearing videos spanning more than 1,500 hours**, a 40-video pilot, 2,145 coded justice claims and 25 conceptions of justice, with initial expert review of 30 segments.[2] Its workflow combines AI extraction, human validation, codebook development and expert review. The Casanare extension examines reparation, restoration, sanction and harm. Combining this foundation and the team's Spanish-language expertise with JEP expert annotations could produce an **expert-annotated benchmark**. LLMs could surface informative comparisons and disagreements, while experts supply judgments and supporting evidence. We would test how this supervision improves conditional encoders, then use retrieval and clustering to explore recurring meanings across the corpus. The SPS team is seeking technical oversight; Columbia Computer Science could strengthen model selection, reproducibility and evaluation while learning from substantive questions and expert feedback. Faculty sponsorship would help shape a bounded study that advances representation learning and supports the preservation and interpretation of victims' accounts for accountability, memory and peacebuilding.

[1] Deshpande et al. C-STS: Conditional Semantic Textual Similarity. EMNLP, 2023.
[2] Tu et al. Linguistically Conditioned Semantic Textual Similarity. ACL, 2024.
[3] Zhang et al. CASE—Condition-Aware Sentence Embeddings for Conditional Semantic Textual Similarity Measurement. EACL, 2026.

[1] Columbia faculty profile; professional biography.

[2] Team-provided *Contexto del Proyecto*, *Script LSA presentation*, and workflow outline, retained in the repository's `sources/` directory. Figures are team-reported.